

DNA extraction protocols used in “Which extraction protocol should I use for my historical freshwater fish samples? A performance analysis on widely used methods”

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1 + 2) DNEASY BLOOD & TISSUE EXTRACTION KIT (1. QIAGEN_OR, 2. QIAGEN_AD)

Reagents: ATL
AL
AW1
AW2
AE / NfW
Prot K
EtOH (96%)

All buffers should be stored at room temperature, Prot K is best stored at 2-8°C. Buffers AL and AW1 contain guanidine salts (do not add bleach).

The membrane of the spin column (QIAamp MinElute column) combines the binding properties of a silica-based membrane with simple microspin technology. This purified DNA typically has an A260/A280 ratio between 1.7 and 1.9 and is 100 bp up to 30 kb in size.

For more detailed info, see the original protocol of Qiagen DNeasy Blood & Tissue Kit.

Protocol: Purification of total DNA from Animal Tissue (spin-column)

- cut up to 25 mg of tissue into small pieces and place into a 1,5 ml reaction tube;
- add 180 µl **ATL** and 20 µl **Protein K**, mix thoroughly and incubate at 56°C until tissue is completely lysed (samples lysed overnight);
optional: after Proteinase K digestion, tissue samples can also be stored in ATL for up to 6 months;
- mix thoroughly (15 sek);
- add 200 µl **AL** and 200 µl **EtOH** (96%), mix thoroughly (a white precipitate may form);
- pipet mixture into the spin column (QIAamp MinElute column) and centrifuge for 1 min (8000 rpm), then discard flow-through and place spin column into a new collection tube;
- add 500 µl **AW1** and centrifuge for 1 min (8000 rpm), then discard flow-through;
- add 500 µl **AW2** and centrifuge for 3 min (14000 rpm), then discard flow-through;
- repeat the last centrifugation step to ensure that any residual EtOH is removed;
- place spin column into final elution tube and add **20 µl NfW**, incubate for at least 1 min (up to 30 min) at room temperature, then centrifuge for 1 min (8000 rpm);
optional: instead of immediate elution, freeze MinElute columns for 24h before final elution;
- repeat elution step into a new elution tube to collect any remaining DNA from the silica-gel membrane of the filter column (elution #2).

3) PHENOL : CHLOROFORM : ISOAMYL-ALCOHOL EXTRACTION (PCI)

Extraction buffer (EB): 10 mM Tris-HCL (pH=7.5)
 10 mM NaCl
 1% SDS (add immediately before use)
 0,5 mg/ml proteinase K (store at -20°C)

For the extraction buffer, a stock solution free of Proteinase K and SDS was prepared in advance. Proteinase K and SDS should then be added only immediately before use. Ideally, all components should be available in liquid form and at the appropriate molarity so that the extraction buffer (EB) can quickly be prepared by mixing them together.

Tris-HCl and NaCl were each prepared at 20 mM so when mixed in equal parts for the EB, they result in a final concentration of 10 mM each.

For an input amount of ~ 50 mg tissue we prepared 300 µl EB per sample. When using the 2 ml Gel Phase Lock tubes for phase separation step, be cautious not to exceed the capacity when adding reagents during washing steps.

Extraction buffer (EB) per sample: 150µl Tris-HCL (20 mM)
 + 150µl NaCl (20 mM)
 + 3 mg SDS (dry powder)
 = **300 µl EB**

Other reagents per sample: + 12 µl DTT (1M)
 + 10 µl proteinase K (600mAu/ml)
 + ~50 mg tissue

PCL = Phenol : Chloroform : Isoamyl Alcohol (25 : 24 : 1)

CL = Chloroform : Isoamyl Alcohol (24 : 1)

Pipette tips were cut beforehand with a heated (sterile) blade to avoid shearing of DNA during pipetting. The modified tips were then be UV-irradiated. Unless stated otherwise, all centrifugation steps were performed at room temperature and maximum speed.

PCI protocol for DNA isolation

- cut up to 50 mg of tissue into small pieces and place into a 1,5 ml reaction tube;
- add 300 µl EB, 12 µl 1M DTT and 10µl proteinase K, mix thoroughly and incubate at 45°C until tissue is completely lysed (samples lysed overnight);
- wash the isolate with PCI (**3x**):
 - add 1050 µl PCI and slowly invert tube until an emulsion forms;
 - incubate for 15 min at room temperature, then centrifuge for 15 min;
 - transfer the upper phase to a Gel Phase Lock Tube and repeat the washing steps twice;
- wash the isolate with CL (**1x**):
 - add the same volume CL as used for EB (~ 300µl) and slowly invert tube;
 - incubate 5 min at RT, then centrifuge for 5 min;
 - transfer the upper phase to a new reaction tube (1,5 ml);

- add 1/10 EB- volume (~ 30µl) **3M NaAc** (pH=5.2) and twice the EB-volume (~600µl) **70% EtOH** and freeze for at least **1h at -20°C**;
- centrifuge for **10 min**, then slowly pour the EtOH and let the DNA pellet dry (at the bottom of the tube);
optional: leave the tube at 37°C for a few minutes (open lid);
optional: repeat the wash step with EtOH;
- elute DNA in 50µl NfW;
optional: incubate at 50°C overnight for better elution.

4) FIRST-MAGNETIC FORENSIC KIT (GEN-IAL)

Reagents: Lyse
Enzyme (proteinase K)
Magnetic particle solution (MPL)
Wash 1
Wash 2

Store solutions in a dark place, the magnetic particles (MPL) need to be stored upright (beads should always be covered by the buffer, never allow to freeze or dry out), make sure that solution is homogeneous before usage (briefly heat or shake).

Manual DNA isolation protocol:

- cut up to 25 mg of tissue into small pieces and place into a 1,5 ml reaction tube;
- add 20 µl proteinase K and 500 µl Lyse (the tissue sample should at least be covered by buffer);
- incubate at 56°C overnight;
- centrifuge for 5 min (10.000 g) and transfer the supernatant into a new reaction tube (or remove any tissue remnants with tweezers);
- add 20 µl MPL and 250 µl EtOH (96%), mix thoroughly and incubate for 2 min at room temperature;

DNA purification protocol (magnetic rack):

- place reaction tube into the magnetic rack and wait for at least 1 min, the pipet off supernatant;
- add 800 µl **Wash**, remove tube from magnetic rack and mix thoroughly by hand (min. 5 sek);
- place reaction tube back to the magnetic rack and wait for at least 1 min, then pipet off supernatant;
- add 400 µl **Wash 2**, remove tube from magnetic rack and mix thoroughly by hand (min. 5 sek);
- place reaction tube back to the magnetic rack and wait for at least 1 min, then pipet off supernatant;
- add 20 µl **NfW** and resuspend (MPL should be completely resolved);
- remove the tube from the magnetic rack and incubate for **10 min at 56°C** (while incubating mix from time to time);
- to get rid of the MPL, place reaction tube to the magnetic rack and wait for at least 1 min, then transfer the DNA to a new (final) reaction tube.

5) TISSUE AND HAIR EXTRACTION KIT (FOR USE WITH DNA IQ) (PROMEGA)

Description: this protocol uses paramagnetic resin to purify DNA. although this kit contains strong denaturing agent, some samples (e.g. tissue, hair) require pretreatment with proteinase K to ensure complete lysis. DNA can then be purified using DNA IQ System. Be careful, if sample is contaminated with microbes, the microbial DNA also will be isolated. DNA molecules less than 80 bp in length (PCR inhibitors) are selectively removed.

<u>Extraction buffer (EB):</u>	incubation Buffer	386 µl
	proteinase K (18mg/ml)	10 µl
	<u>1-Thioglycerol*</u>	<u>4 µl</u>
	total volume per sample	400 µl

* add 1 µl 1-Thioglycerol for every 100 µl of Lysis Buffer; mix by inverting several times

Protocol: DNA extraction via the Maxwell system (Promega)

- cut up to 25 mg of tissue into small pieces and place into a 1,5 ml reaction tube;
- add 400µl freshly prepared EB and incubate at 56°C overnight;
- transfer of the lysate (~400µl) into a new tube and add 200µl Lysis buffer, mix thoroughly;
- place the cartridges (Maxwell FSC DNA IQ Casework Cartridge) to the deck tray, press down and peel back the seal;
- transfer the pre-processed samples to well #1 of the cartridges and mix by using a pipette tip;
- place a plunger into well #8 of each cartridge;
- place an empty elution tube (from the kit) in the deck tray and add 50µl NfW to the bottom of it.

6) DABNEY & ROHLAND DNA EXTRACTION PROTOCOL (D&R)

Reagents: extraction buffer (EB, freshly prepared)
binding buffer (Dabney buffer) (BB, can be stored up to one month)
TET buffer (can be stored long-term)
PE buffer

Prepare 1 ml extraction buffer per sample freshly on the day of extraction:

reagents EB	final concentration	amount for 1 ml
GuSCN powder pure	5 M	0,59 g
5 M NaCl	25 mM	5 µl
1M Tris-HCL, pH 8.0	50 mM	50 µl
0.5 M EDTA, pH 8.0	20 mM	44,4 µl
10 % Tween-20	1 %	100 µl
β-mercaptoethanol	1 %	10 µl
HPLC H ₂ O	-	up to 1 ml

The binding buffer (BB) can be stored up to 1 month. There may be difficulties in dissolving the guanidine powder (GuSCN), make sure that guanidine is completely dissolved and that no lumps remain in the final buffer. Note: per sample a volume of 13 ml BB is required.

reagents BB	final concentration	amount for 50 ml
Guanidine Hydrochloride (MW 95,53)	5 M	23,9 g
Isopropanol	40%	20 ml
10% Tween-20	0,05%	250 µl
Sodium acetate 3M solution	90 mM	1,5 ml
HPLC H ₂ O	-	up to 50 ml

The TET buffer can be stored long-term, per sample a volume of 25µl is required (for two elutions).

reagents TET	final concentration	amount for 50 ml
1 M Tris-HCl solution, pH 8.0	10 mM	500 µl
0.5 M EDTA solution	1 mM	100 µl
10% Tween-20 solution	0,05%	250 µl
dH ₂ O	-	up to 50 ml

Binding apparatus: Zymo-spin V column extension reservoir (remove spin column)
Qiagen MinElute silica spin column
50 ml Falcon tube

Assemble the binding apparatus by pressing the MinElute column firmly onto the extension reservoir of the Zymo-spin (with some force, it can take a bit of effort). Remove the MinElute collection tube, then cut off the cap and save it for later (e.g., put it directly on the collection tube to keep it sterile). Both the MinElute (side) and Falcon tubes (lid) should be labelled to ensure samples are not mixed up. Next place the reservoir (including MinElute) into the Falcon tube—it should fit well, but it is necessary to secure the Falcon tube cap with sticky tape before centrifuging to avoid spilling.

Dabney & Rohland protocol for DNA extraction

- cut up to 50 mg of tissue into small pieces and place into a 2 ml reaction tube;
- add 1 ml **EB** (extraction buffer), mix thoroughly and incubate at 37°C with constant rotation for ~18 h (use parafilm to seal the lids);
- centrifuge for 2 min at max. speed (to pellet the tissue remnants at the bottom);

- transfer the supernatant into a fresh reaction tube and mix it with 13 ml **BB** (binding buffer), then pour the mix into the binding apparatus reservoir;
- tape the lid of the falcon tube with sticky tape;
- centrifuge the binding apparatus for 4 min (2500 rpm);
- rotate the binding apparatus for 90° and centrifuge again for 4 min (2500 rpm);
- disassemble the binding apparatus and place the MinElute column in a collection tube and close it with the original lid (retained from earlier);
- centrifuge for 1 min (6000 rpm) to ensure that any residual liquid is removed;
- add 650 µl PE buffer and centrifuge for 1 min (3000 rpm), then discard the flow-through;
- repeat the previous step (add 650 µl PE buffer and centrifuge), then discard the flow-through;
- centrifuge again for 1 min (max. speed) to ensure that any residual liquid is removed, then place the MinElute column into the final elution tube;
- add 12,5 µl TET buffer and incubate for 10 min at room temperature, then centrifuge for 30 sec (max. speed);
- repeat elution step into a new elution tube to collect any remaining DNA from the silica-gel membrane of the filter column (elution #2);